

The Three Genera of D_{IV}^5 — STANDING CONVENTION for the BST paper series (referee-proofing; cite in every paper)

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The Three Genera of D_{IV}^5 — standing convention

0. Purpose

$g = 7$ was mislabeled as a Bergman/genus quantity THREE times on 2026-05-28 (volume constant T2442; “ g = Bergman exponent” in all-5-from- B_2 ; kernel singularity exponent). Same root cause every time: $g = 7$ is the embedding/signature dimension of $SO_0(5,2)$, and it keeps getting confused with a genus. This standing note states the distinction once, for the whole paper series. Every paper invoking “genus,” “Bergman exponent,” or “dimension” cites this note and specifies WHICH quantity it means.

1. The three distinct quantities — ONE genus + Casimir + embedding (CORRECTED)

CORRECTION (2026-05-28 PM): the original v0.1 of this note said “three genera” ($Hua=5$ / $FK=6$ / embedding=7). That itself miscounted — the recheck that this note was meant to prevent caught it. The corrected statement, converged from two independent derivations (Elie Toy 3596: FK multiplicity formula + dimension consistency + convention-free Bergman exponent + Hua kernel; + Keeper multiplicity-formula derivation):

D_{IV}^5 has ONE genus, plus two other distinct quantities that were being mislabeled as genera:

Quantity	Value	What it is	Where it appears
GENUS (FK = Hua = Bergman exponent — ONE genus for Type IV)	$n_C = 5$	complex dimension of D_{IV}^5 ; FK genus $p = 2+(r-1)a+b = 2+3 = 5$ (Type IV: tube/Lorentz cone, $b=0$, $a=n-2$)	Bergman kernel singularity exponent ($K \sim \text{norm}^{\{-5\}}$); c_{FK} measure constant; $\rho_1 = \text{genus}/\text{rank} = 5/2$
CASIMIR	$C_2 = 6$	the quadratic Casimir of B_2	Casimir eigenvalue — NOT a genus
EMBEDDING / SIGNATURE	$g = 7$	$p + q$ of $SO_0(5,2) = 5+2$	signature total — NOT a genus, NOT the kernel exponent

(No $Hua=5$ / $FK=6$ distinction: for Type IV the Faraut-Korányi genus and the Hua kernel exponent agree, both = $n_C = 5$.)

2. The critical rule (corrected)

The intrinsic genus of D_{IV}^5 is 5 — never 6, never 7. $C_2=6$ is the Casimir; $g=7$ is the embedding signature; neither is a genus or the Bergman kernel exponent.

Sanity check (Keeper): the unit ball B^n has Bergman kernel exponent = its genus = $n+1$; the embedding group plays no role. For Type IV_n, the genus = n ($= n_C = 5$ here).

Final lock pending Keeper's primary-source confirmation against the Faraut-Korányi table (Loos as backup). The derivation is solid standard math; the lock is the book-pin — because the genus name flipped three times today from relabeling-from-memory, the standing discipline is: pin to the book, cite the book, stop relabeling from memory.

3. Resolved + open status (as of 2026-05-28 10:50 EDT)

Quantity	Status	Value
Volume constant c_{FK} (T2442)	RESOLVED — STANDS (Grace provenance)	computed at Hua genus = 5; $c_{FK} = 225/\pi^{9/2}$, $9/2 = 5 - 1/2$
g in “ g = Bergman exponent” (all-5-from- B_2)	RESOLVED	$g = 7$ = embedding/signature, NOT Bergman exponent; n_C = Hua genus is the Bergman-relevant one
Kernel singularity exponent	OPEN → near-resolved	Elie numerical kernel ν ; Toy 3579 read $\nu = 5$ (Hua) → exponent/rank = $5/2 = \rho_1$; awaiting explicit confirmation (Keeper routed to Elie)

4. Where $g = 7$ LEGITIMATELY appears

$g = 7$ is a genuine BST primary with legitimate roles — just NOT as a genus: - Embedding/signature dimension $p+q = 5+2$ of $SO_0(5,2)$ (its definition) - Mersenne prime $M_{\{N_c\}} = 2^{\{N_c\}} - 1 = 7$ (arithmetic overdetermination; cyclotomic chain) - $n_C + \text{rank} = 5 + 2 = 7$ (signature decomposition) - BST primary in observable formulas (Serre long-root coefficient $N_c \cdot g = 21$; Weinberg $g/(\text{rank}+g)$; etc.)

These are all correct. The ONLY error is calling g a genus or Bergman exponent.

5. Standing instruction for the paper series

Every BST paper (A1-A4, B1-B6, C1-C4, Vol 0-16) that invokes a genus, Bergman exponent, dimension, or kernel power: 1. Cite this note 2. State WHICH of the three quantities (Hua genus 5 / FK genus 6 / embedding 7) it uses 3. Never write “Bergman exponent = g/rank ” without confirming it's the genus, not the signature

This single convention prevents the recurring g -mislabel recheck cycle permanently.

6. Provenance

- Lyra April-10 genus note: already had all three conventions correct (Hua=5, FK=6, embedding=7)
- 2026-05-28 Thursday: three mislabels caught + resolved (T2442 stands; all-5-from- B_2 g =signature; kernel exponent routed to Elie)
- Keeper recommendation (10:48): make it a paper-series standing convention
- Filed in A1 v0.2 §1 + Vol 16 outline §1.5 + this series-wide note

— Lyra, three-genera standing convention v0.1 filed (Keeper recommendation). Hua genus = $n_C = 5$ (Bergman/kernel/volume); FK genus = $C_2 = 6$ (Casimir); embedding/signature = $g = 7$ (NOT a genus). Intrinsic quantities are never $g=7$. Every paper cites this + specifies which. Prevents the recurring g -mislabel recheck cycle permanently. Kernel exponent value ($5/2$ Hua, per Elie 3579 $\nu=5$) pending Elie explicit confirmation.